Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

# SDC Hackathon

# Technical Tools Declaration

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Submission Deadline: [Insert date]**

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

***Instagram:*** @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

# 1. TEAM AND PROJECT OVERVIEW

## Team Name

tudou

## Project / Solution Name

Pivott

## Name (Leader)

Ho Xiang Ting

## Email & Phone (Leader)

aneoho@gmail.com | 013-368 0236

## University Affiliation(s)

Xiamen University Malaysia

## Live Prototype URL

https://pivott.replit.app

## Code Repository URL (public / access granted to judges)

https://github.com/truffleloaf/Tudou-Pivott

# 2. ARCHITECTURE & TECHNOLOGY STACK

## 2.1. System Architecture Overview

*Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.*

***[Maximum 200 words]***

A guest uses the Next.js/React browser interface. Profile choices and progress remain in device storage. The TypeScript engine calculates bounded task exposure, role fit and a learning plan from synthetic demo or dated public snapshots. Next.js route handlers serve catalogue and job data. A separate, optional FastAPI/SQLModel service mirrors the scoring contract and supports imports and PostgreSQL checks; the public Vercel app does not require it. If sign-in is configured, Supabase Auth issues a token and owner-bound row-level security protects the saved profile. The public-data refresh workflow imports O*NET 31.0 and Arbeitnow into reviewable snapshots. Vercel hosts the main app; SDC sponsor-provided Replit hosts a second public instance used for runtime debugging. Neither public instance calls a GenAI model to score users. AI systems were development tools only: a coding agent drafted code, Claude Sonnet 5.5 helped debug, and ChatGPT/Codex assisted documentation and slides. Human developers reviewed and tested the outputs.

# 2. ARCHITECTURE & TECHNOLOGY STACK

## 2.2. Languages, Frameworks & Libraries

*List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.*

| Name & Version | Purpose in Project | License |
|---|---|---|
| Next.js 16.4.0 | Public web app and API routes | MIT |
| React 19 | Interface components | MIT |
| TypeScript 5.7.3 | Typed frontend and scoring engine | Apache-2.0 |
| Tailwind CSS 4.3.3 | UI styling | MIT |
| FastAPI >=0.115 | Optional Python API | MIT |
| Supabase JS ^2.118 | Optional Auth/profile sync | MIT |

# 3. APIs & CLOUD SERVICES

## 3.1. External APIs

*List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.*

| API and Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
|---|---|---|
| Arbeitnow API | Public job-listing refresh | Public/free |
| Supabase Auth API | Optional user sign-in/profile sync | Account; tier not confirmed |
| GenAI inference API | None in running product | Not applicable |
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# 3. APIs & CLOUD SERVICES

## 3.2. Cloud Platforms & Deployment

*List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.*

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| Replit | Official SDC sponsor deployment; hosting and debugging | Sponsor access; region not supplied |
| Vercel | Main Next.js hosting and consent-gated analytics | Plan/region not confirmed |
| Supabase | Optional Auth and profile Postgres | Project plan/region unverified |
| GitHub / Actions | Code repository, CI and data refresh | Public repo; hosted CI |
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# 4. AI TOOLS, MODELS & DATASETS

## 4.1. AI Models Used in the Solution

*List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.*

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
| None | Not applicable | Deterministic formulas; no LLM in app | Not applicable |
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# 4. AI TOOLS, MODELS & DATASETS

## 4.2. Datasets

*List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.*

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| | | | | |
| O*NET 31.0 | onetcenter.org | Public | CC BY 4.0 | No |
| Arbeitnow jobs | arbeitnow.com | Public dated feed | Provider terms | No |
| UN SDG names | sdgs.un.org | Public | UN source cited | No |
| Aisyah demo | Team-created | Synthetic | Team-created | No |
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# 4. AI TOOLS, MODELS & DATASETS

## 4.3. AI-Assisted Development Tools

*Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.*

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

| Tool & Version | Used For | Extent of Use (minor / moderate / major) |
|---|---|---|
| Model-API coding agent; model unrecorded | Backend code/tests and first drafts | Major |
| Claude Sonnet 5.5 | Debugging suggestions checked by team | Moderate |
| ChatGPT / Codex; model varies | Research, pitch deck and submission text | Moderate |
| Cursor + Grok 4.6 | Main IDE: scan code, inspect localhost | Major as IDE; AI extent unquantified |

# 4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

***[Maximum 100 words]***

The backend developer wrote the API and data-model specifications and fixed expected outputs before agent tasks. The team reviewed generated diffs, reran frontend and backend tests, compared Python and TypeScript scoring against canonical values, and checked error and privacy boundaries. The developer repaired a stale PostgreSQL verifier and CI job, resolved deployment routing and reviewed pull requests before merging. Claude's debugging suggestions were treated as hypotheses and checked against code, logs and observed behavior. Cursor and Replit were used to inspect localhost and deployed behavior. The team owns the final code, claims and demo.

# 5. SPONSOR-PROVIDED RESOURCES

*Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.*

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

| Sponsor | Resource (API / credits / platform / data) | How It Was Used in the Project |
|---|---|---|
| Replit | Sponsor-provided hosting platform | Public deployment and debugging |
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# 5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

***[Maximum 100 words]***

Official SDC sponsor-provided Replit was used to host the public https://pivott.replit.app instance and debug runtime behavior. It is not essential to the deterministic scoring engine: the Next.js application also has a Vercel deployment. If Replit became unavailable, the team could continue the core demo from Vercel, subject to checking the current deployment and links. No sponsor-provided GenAI inference API is used by the running product.

# 6. SECURITY, PRIVACY & RESPONSIBLE USE

## 6.1. Data Handling & Privacy

*What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.*

***[Maximum 150 words]***

Guests can keep role choices, task/skill ratings, selected routes and plans in browser storage; these remain until the user clears or replaces them. A resume is read locally in the browser and not sent to the catalogue refresh or analytics service. Optional Supabase sign-in processes email and authentication data and can sync the saved profile to an owner-bound database row; the account copy remains until deletion. The public job-feed refresh does not send profile data to Arbeitnow. Vercel Analytics is consent-gated; the Replit version currently says anonymous analytics are not enabled. No profile is sent to a GenAI inference API for scoring. Provider operational logs follow provider retention terms; live account configuration must be verified before promising sign-in to judges.

# 6. SECURITY, PRIVACY & RESPONSIBLE USE

## 6.2. Known Limitations & Risks

*Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.*

***[Maximum 150 words]***

O*NET is US occupation data and the Arbeitnow feed is mainly German, so neither establishes Malaysian demand. A single dated posting snapshot cannot prove a trend; the app shows insufficient data where history is thin. Exposure, fit, confidence and learning hours are team-model estimates, not job-loss probabilities or hiring decisions. Demo profiles and illustrative signals are synthetic. Missing vetted learning resources remain explicitly unavailable. Optional Supabase account configuration and live email delivery have not been independently verified in this workspace. The public API has no application-level authentication or rate limiter; hosting-edge controls are needed before broad use. Mitigations are source/date labels, bounded formulas, scenario comparison, local-first guest use, separate demo/live data, tests and human review. Future work should validate results with Malaysian workers and local data.

# 7. COMPLIANCE & READINESS

*Tick all boxes that apply before submitting.*

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.

- [x] Any use of official SDC sponsor resources has been declared in Section 5.

- [x] The prototype described here is accessible to judges through the links provided in Section 1.

- [x] All fields in this template have been completed within the word limits stated.

# 8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]

25/10/2026